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Door stop mechanism

Abstract

An example door stop mechanism is provided. The door stop mechanism includes a body, a connection member, a spring, and a brake. The spring is coupled to the body and the connection member. The brake is positioned adjacent to the body. The brake moves between an engaged position via a spring force of the spring that causes the body to interact with the brake such that the brake contacts a track of an overhead door and an disengaged position via a force that acts against the spring force to pull the body away from the brake such that the brake moves away from the track to allow the overhead door to move.

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Background/Summary

BACKGROUND

(1) Overhead doors to cover external and internal openings may comprise panel sections that are connected by hinges. An overhead door can be raised into an open position and lowered to a closed position through rollers in a track system. The track system has a vertical section mounted to a wall, and a horizontal section mounted to a ceiling or overhead structure. In addition to the track

system, a cable or strap system interacts with a counterbalance system comprising a torsion spring bar. The torsion spring bar comprises a torsion spring and cable drums located on each end of the torsion spring bar. The cable drum is connected to a first end of a cable or strap, and the second end of the cable or strap is attached to a bottom bracket, which is mounted to the bottom-most panel of the door.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of an overhead door with an example door stop mechanism of the present disclosure;
- (2) FIG. 2 is an isometric view of an example door stop mechanism in a disengaged position of the present disclosure;
- (3) FIG. 3 is an isometric view of the example door stop mechanism in an engaged position of the present disclosure;
- (4) FIG. 4 is an exploded view of the example door stop mechanism of the present disclosure;
- (5) FIG. 5 is an side view of a bracket of the example door stop mechanism of the present disclosure;
- (6) FIG. 6 is an isometric view of the example door stop mechanism of the present disclosure on a panel of the overhead door relative to a track and in in a disengaged position;
- (7) FIG. 7 is an isometric view of the example door stop mechanism of the present disclosure on a panel of the overhead door relative to a track and in in an engaged position;
- (8) FIG. 8 is an isometric view of an example gear box with a member that is attached to a strap connected to the example door stop mechanism of the present disclosure;
- (9) FIG. 9 is a view that illustrates how the door stop mechanism is coupled to a door; and
- (10) FIG. 10 is an exploded view of another embodiment of the door stop mechanism.

DETAILED DESCRIPTION

(11) Examples described herein provide examples of a door stop mechanism for an overhead door. As discussed above, an overhead door can include a cable or strap system to help open and close the panels of the overhead door. One issue with cable or strap systems for overhead doors is that the tension of the cable or strap should be consistent so as not to cause misalignment or prohibit movement of panels of the overhead door when opened or closed.

(12) In some instances, when the door is closing, there may be a point where the door is in a vertical section of the track and moved by gravity. Only the strap being controlled by the torsion spring bar controls the descending velocity of the overhead door. An uncontrolled descending velocity of the moving panels of the overhead door may be undesirable, and reliable means to arrest the movement of the overhead door is therefore desired.

(13) The present disclosure provides a door stop mechanism to stop movement of the overhead door when tension in the strap is lost. For example, while tension in the strap is maintained, the door stop mechanism may be disengaged to allow the door to move freely within the track system. When tension in the strap is lost, the door stop mechanism may be moved into an engaged position to prevent the door from moving until the strap is replaced or tension in the strap is restored.

(14) In an example, the door stop mechanism may include a braking system that engages the track. The braking system may use friction to “catch” the track and prevent movement of the overhead door.

(15) The door stop mechanism of the present disclosure may provide a low-cost solution to arresting movement of overhead doors when tension in the strap is lost. The door stop mechanism of the present disclosure may also be retrofitted to existing overhead doors and provide a relatively low profile and clean design.

(16) FIG. 1 illustrates an interior front view of an overhead door system **100** with door stop mechanism **118** of the present disclosure. FIG. 1 illustrates a view of the overhead door system **100** with a door **102** in the closed positioned.

(17) In one embodiment, the overhead door system **100** includes the door **102**. The door **102** may be comprised of a plurality of panels **104.sub.1** to **104.sub.n** (hereinafter also referred to individually as a panel **104** or collectively as panels **104**). The panels **104** may be vertically arranged. In some embodiments, the panels **104** may be movably coupled via panel fasteners or hinges (not shown) or may be disconnected from one another.

(18) The panels **104** may be constructed of the same materials or different materials. In one embodiment, each panel **104** may be a metal frame structure without a middle panel (e.g., an open panel). The metal frame structure may comprise at least two vertical stiles and at least two horizontal stiles connected at right angles. The metal frame may further comprise a third vertical stile, a fourth vertical stile, a fifth vertical stile, and a sixth vertical stile forming multiple middle areas defined by the metal frame structure. The middle areas of this metal frame can be open, may comprise a solid metal panel, may comprise an open metal structure (e.g., screen, grid, grate, woven metal structure, or the like), may comprise a polyacrylate panel that is clear or translucent, may comprise polymethylmethacrylate (PMMA) that is clear or translucent, may comprise a polypropylene panel that is clear or translucent, may comprise a glass panel that is clear or translucent, or may comprise a laminate structure that is intended to have impact resistance.

(19) In one embodiment, the overhead door system **100** may include a track system that includes a first track **108.sub.1** and a second track **108.sub.2** that are oriented parallel to each other and located adjacent to the ends of each panel. The door **102** may be guided into an open and closed position via the first track **108.sub.1** and the second track **108.sub.2**. For example, wheels **106** or other mechanical means (not shown) may be coupled to the ends of each panel **104**. The wheels **106** may be fitted inside of the first track **108.sub.1** and the second track **108.sub.2**. As the door **102** is opened and closed, the wheels **106** may travel along the first track **108.sub.1** and the second track **108.sub.2**.

(20) In one embodiment, the overhead door system **100** may include a shaft **110** that is located over the door **102**. The shaft **110** may be coupled to a gear box **114** on one or both ends of the shaft **110**, and to a motor **112**. The gearbox **114** may be coupled to an end of the shaft **110**. The motor **112** may cause the shaft **110** to rotate and assist in the movement of the door **102** between an open position and a closed position. A drum or spool may be coupled to the ends of the shaft **110** or coupled to the gear box **114**.

(21) In one embodiment, a connection member **116** may be coupled to a portion of the spool or drum (shown in FIG. 8) and the door stop mechanism **118**. The connection member **116** may be a strap or a cable. A strap may be fabricated from fabric or metal and may have a flat or ribbon shape. A cable may be fabricated from steel and have a cylindrical shape.

(22) In one embodiment, FIG. 1 illustrates an example where the connection member **116** is a strap (hereinafter also referred to as the strap **116**). The strap **116** may be oriented such that a width of the strap **116** is parallel to a width of the track **108.sub.1** and/or **108.sub.2**. The drum may be selected to be concentrically aligned with the shaft **110**, allowing for the spooling of the strap **116**, or may be a gear box that allows for the strap to be oriented parallel to a width of the track **108.sub.1** and/or **108.sub.2**. Further details of how the strap **116** is coupled to the drum and the door stop mechanism **118** are discussed in further detail below.

(23) Although a single door stop mechanism **118** and single strap **116** are illustrated in FIG. 1, it should be noted that the overhead door system **100** may include two door stop mechanisms **118** and two straps **116**. For example, a door stop mechanism **118** and a strap **116** may be deployed on each end of the door **102**.

(24) In one embodiment, the doorstop mechanism **118** may be positioned only on a bottom most panel **104.sub.1**. As discussed in further detail below, the tension in the strap **116** may control

whether the door stop mechanism **118** is engaged or disengaged. When tension in the strap **116** is lost, the door stop mechanism **118** may move to an engaged position that arrests movement of the door **102** or arrests movement of the panels **104** within the tracks **108.sub.1** and **108.sub.2**. When tension in the strap **116** is maintained, the door stop mechanism **118** may move to a disengaged position that allows door **102** to move freely within the tracks **108.sub.1** and **108.sub.2**.

(25) In one embodiment, the strap **116** may be tightened to have an initial amount of tension to keep the door stop mechanism **118** in a disengaged position. Tension in the strap **116** may be lost when the amount of tension becomes less than the initial amount of tension that was set in the strap **116**. Tension may be lost in the strap **116** over time due to stretching of the strap **116**, movement or slight separation of the panels **104**, misalignment of the door **102** within the tracks **108.sub.1** and **108.sub.2**, and the like.

(26) FIG. 2 illustrates an isometric view of an example door stop mechanism **118**. In one embodiment, the door stop mechanism **118** may include a strap connection member **204**, a body **206**, and a brake **208**. The strap connection member **204**, the body **206**, and the brake **208** may be coupled together mechanically via a combination of protruding members **218**, **219**, and **224** inserted into slots **226**, **220**, and **222** and a spring **230**. The strap connection member **204**, the body **206**, and the brake **208** may be coupled to a bracket **202**.

(27) In one embodiment, bracket **202** may include a first side **232** and a second side **234**. The first side **232** and the second side **234** may be coupled together at a right angle, perpendicularly, or at approximately 90 degrees. The first side **232** and the second side **234** may be formed from a single piece of metal or sheet metal. The first side **232** may include one or more openings **236.sub.1** to **236.sub.m** to couple the bracket **202** to a bottom most panel **104.sub.1** of the door **102**, illustrated in FIG. 1. A mechanical fastener (e.g., a screw, a nut and bolt, a pin, and the like) may be inserted into each of the openings **236.sub.1** to **236.sub.m** and into corresponding openings (not shown) in the bottom most panel **104.sub.1** to secure the bracket **202** to the bottom most panel **104.sub.1**.

(28) In one embodiment, the body **206** may be coupled to the second side **234** of the bracket **202**. For example, the second side **234** may include the protruding members **218** and **219**. The protruding members **218** and **219** may each have a cylindrical shape with a cylindrical body that has a smaller diameter than the cylindrical head. The body **206** may include slots **220** and **222**. The slots **220** and **222** may include a first portion that matches a diameter of the cylindrical heads of the protruding members **218** and **219**. The slots **220** and **222** may include a second portion that is smaller in diameter or width than the first portion that matches the diameters of the cylindrical bodies of the protruding members **218** and **219**.

(29) The protruding member **218** may be inserted into the slot **220**, and the protruding member **219** may be inserted into the slot **222**. The cylindrical bodies of the protruding members **218** and **219** may move within the second portions of the slots **220** and **222**. The larger diameters of the cylindrical heads of the protruding members **218** and **219** may keep the bodies **206** secured against the second side **234** of the bracket **202**.

(30) The body **206** may include a protruding member **224**. The protruding member **224** may have similar dimensions and features to the dimensions and features of the protruding members **218** and **219** (e.g., the cylindrical body and a larger cylindrical head). The strap connection member **204** may include a slot **226**. The slot **226** may have a similar size and similar features to the sizes and features of the slots **220** and **222** (e.g., a first portion with a larger diameter and a second portion with a smaller diameter).

(31) The strap connection member **204** may be coupled to the body **206** by inserting the protruding member **224** through the slot **226**, similar to how the protruding members **218** and **219** are inserted into the slots **220** and **222**. The strap connection member **204** may be secured to the body **206** via a spring **230**. The spring **230** may be coupled to the protruding member **224** and to the protruding member **218** via a screw or other type of mechanical fastener.

(32) In one embodiment, the protruding members **218**, **219**, and **224** may be aligned vertically on

the second side **234** of the bracket **202**. In other words, centers of the protruding members **218**, **219**, and **224** may lie on a common vertical line that would run parallel to an edge of the second side **234**. The amount of movement of the strap connection member **204** and the body **206** may be defined by lengths of the slots **220**, **222**, and **226**.

(33) In one embodiment, the spring **230** may be loaded to pull the strap member **204** downwards. In other words, the spring **230** may be loaded to move the strap connection member **204** closer to the protruding member **218**.

(34) The strap connection member **204** may also include a slot **228**. The strap **116** may be fed through the slot **228**. As noted above, when tension is applied to the strap **116**, the tension in the strap **116** may act against the spring **230** and move the strap connection member **204** in an upward direction. In other words, the tension in the strap **116** may pull the strap connection member **204** away from the protruding member **218**. As a result, the bottom of the slot **226** may pull up against the protruding member **224** on the body **206** and move the body **206** upwards. The upward movement of the body **206** may move the brake **208** into a disengaged position.

(35) When the tension in the strap **116** is lost, the force of the spring **230** may overcome the remaining tension in the strap **116** and cause the strap connection member **204** to move towards the protruding member **218**. This causes the slot **226** to move lower, thereby allowing the slots **220** and **222** to also move lower against the protruding members **218** and **219**, respectively. This may cause the body **206** to move in a downward direction. The downward movement of the body **206** may move the brake **208** into an engaged position. FIGS. 3, 6, and 7 illustrate further details of how the brake **208** is moved between the disengaged and engaged position.

(36) In one embodiment, the brake **208** may be rotatably coupled to the second side **234** of the bracket **202**. The brake **208** may include a pivot member **214** that is inserted into an opening on the second side **234** (shown in FIG. 4). The brake **208** may be secured via a torsion spring **214** that is wrapped around the pivot member **214**. The torsion spring **214** may be loaded to cause the brake **208** to move away from the external surface of the track **108.sub.1** or **108.sub.2** into a disengaged position.

(37) In one embodiment, the brake **208** may include an arm **242** that is coupled to the pivot member **214**. The arm **242** may interact with a cam surface **240** of a cam **238** of the body **206**. The cam surface **240** may be inclined or formed at an angle to properly interact with the arm **242**, as illustrated in FIG. 5, and discussed in further detail below. As the body **206** moves upward and downward, the cam surface **240** may move away from the arm **242** or move against the arm **242**. When the cam surface **240** does not contact the arm **242**, the torsion spring **214** may cause the pivot member **214** and the arm **242** to rotate clockwise and to move the brake **208** into a disengaged position. When the cam surface **240** acts upon the arm **242**, the cam surface **240** may apply a force that overcomes the force applied by the torsion spring **214**. This may cause the pivot member **214** to rotate in a counter clockwise direction as the cam surface **240** moves further downward against the arm **242**. The counter clockwise rotation of the pivot member **214** may cause the brake **208** to move to an engaged position.

(38) In one embodiment, the brake **208** may have a curved outer braking surface **210**. The surface **210** may be curved along a single axis of the surface. For example, the curved outer braking surface **210** may be formed by curving a rectangular surface around a horizontal axis or around the pivot member **214**. Said another way, the curved outer braking surface **210** may appear as a section cut away from a curved surface of a cylinder.

(39) The curved outer braking surface **210** may face the tracks **108.sub.1** and/or **108.sub.2**. The curved outer braking surface **210** may include a plurality of contact points **212.sub.1** to **212.sub.n** (hereinafter also referred to individually as a contact point **212** or collectively as contact points **212**). The contact points **212** may provide multiple surfaces that can “catch” the surface of the track **108.sub.1** or **108.sub.2**. Thus, the multiple contact points **212** ensure that the brake **208** can engage the surface of the track **108.sub.1** or **108.sub.2**. The curved outer braking surface **210** allows more

contact points **212** to gradually engage the surface of the tracks **108.sub.1** or **108.sub.2** to gradually stop the downward movement of the door **102**.

(40) In one embodiment, the contact points **212** may be formed with a series of alternating peaks and valleys or depressions. For example, the contact points **212** may be formed as a series of teeth or in a saw-tooth pattern. However, any pattern may be deployed for the contact points **212** that can be applied to the curved outer braking surface **210**. For example, the contacts points **212** may be formed as a file textured surface with a cross-hatch pattern, or any other type of irregular surface that may create friction to help stop downward movement of the door **102**.

(41) FIG. **3** illustrates an example of movement of the strap connection member **204**, the body **206**, and the brake **208** when the door stop mechanism **118** is moved to an engaged position. For example, when tension is lost on the strap **116**, the force of the spring **230** may pull the body **206** downward in a direction shown by an arrow **304** towards the protruding member **218** that is in a fixed position on the second side **234** of the bracket **202**.

(42) As the body **206** moves downward, the cam surface **240** of the cam **238** may begin to act upon the arm **242** of the brake **208**. The cam surface **240** may be cut an angle to form an inclined surface that can cause the arm **242** to rotate around the pivot member **214** as shown by arrow **302**. The force of the cam surface **240** acting upon the arm **242** may be greater than the force applied by the torsion spring **216** to cause the arm **242** to rotate.

(43) The rotation of the arm **242** may move the brake **208** towards a surface of the track **108.sub.1** or **108.sub.2** in a direction shown by an arrow **306**. The contact points **212** of the curved outer braking surface **210** may gradually begin to interact with the surface of track **108.sub.1** or **108.sub.2**. The contact points **212** may “grab” or “catch” the surface of the track **108.sub.1** or **108.sub.2** to arrest or stop the downward movement of the door **102**.

(44) When tension in the strap **116** is restored, the force applied by the tension in the strap **116** may act against the spring **230** to pull the strap connection member **204** upward in a direction opposite the arrow **304**. The upward movement of the strap connection member **204** may pull the body **206** in the upward direction such that the cam surface **240** moves away from the arm **242** of the brake **208**. As a result, the force applied by the torsion spring **216** may allow the arm **242** to rotate in a clockwise direction that is opposite the arrow **302**. The rotation of the arm **242** in the clockwise direction may move the brake **208** away from the surface of the track **108.sub.1** or **108.sub.2** in a direction opposite the arrow **306** into a disengaged position.

(45) FIG. **4** illustrates an example exploded view of the door stop mechanism **118**. FIG. **4** illustrates the bracket **202** comprising the first side **232** and the second side **234**. As shown in FIG. **4**, the second side **234** of the bracket **202** may include protruding members **218** and **219**. FIG. **4** also illustrates an opening **254** on the second side **234** of the bracket **202**.

(46) The pivot member **214** may be inserted through the opening **254** and held by a clip **256**. The torsion spring **216** may be looped around the pivot member **214** and held in place by a clip **260**.

(47) The body **206** may be coupled to the second side **234** of the bracket by inserting the protruding member **218** through the slot **220** and the protruding member **219** through the slot **222**. The strap connection member **204** may be coupled to the body **206** by inserting the protruding member **224** through the slot **226**. The spring **230** may be secured between the protruding member **224** and the protruding member **218**. For example, a first screw **250** may be fed through a first loop on a first end of the spring **230** and into a threaded opening of the protruding member **224**. A second screw **252** may be fed through a second loop on a second end (opposite the first end) of the spring **230** and into a threaded opening of the protruding member **218**. The threaded opening may be located in a center of the protruding member **224** and **218**.

(48) FIG. **5** illustrates a close up view of the cam **238** and the cam surface **240**. As noted above, the cam surface **240** may be cut or machined to have an inclined surface. In one embodiment, the inclined surface may be formed to have an angle **504**. The angle **504** may be measured relative to an axis **502** that runs parallel to an edge **508** of the body **206**.

(49) In one embodiment, the angle **504** of the cam surface **240** is set such that the body **206** may move smoothly in a downward direction. In addition, the angle **504** of the cam surface **240** may be set such that the inclined surface of the cam surface **240** gradually acts against the arm **242** of the brake **208** to rotate the arm around the pivot member **214**.

(50) In one embodiment, the angle **504** may be greater than 90 degrees. In one embodiment, the angle **504** may be from about 155 degrees to 175 degrees. In one embodiment, the angle **504** may be approximately 165 degrees.

(51) FIGS. **6** and **7** illustrate more detailed views of the brake **208** in a disengaged and in an engaged position, respectively. FIG. **6** illustrates the brake **208** in a disengaged position. In one embodiment, the track **108.sub.1** may include a panel **604** that can be coupled to the wall of an opening where the overhead door system **100** is installed. The track **108.sub.1** and/or track **108.sub.2** may include a surface **602** that interacts with the brake **208** and a wheel track **606**. The wheel **106** of the panel **104.sub.1** may travel within the wheel track **606** as the door **102** moves between the open and closed positions.

(52) In the disengaged position, the brake **208** moves away from the surface **602**. In other words, tension on the strap **116** causes the body **206** to move away from the brake **208** and allows the torsion spring **216** to rotate the brake **208** such that the brake **208** does not contact the surface **602**.

(53) FIG. **7** illustrates the brake **208** in the engaged position. When tension on the strap **116** is lost, the spring **130** may cause the body **206** to move downward against the brake **208**. The brake **208** may rotate towards the surface **602** as the cam surface **240** acts against the arm of the brake **208**. The contact points **212** of the brake **208** may engage or contact the surface **602** to stop movement of the door **102**. Further details of how the door stop mechanism is coupled to the door **102** are illustrated in FIG. **9**, and discussed below.

(54) FIG. **8** illustrates a more detailed view of the gear box **114** with a drum **802**. In one embodiment, to provide a clean look, the strap **116** may be positioned such that a width **804** of the strap **116** is parallel to a width **806** of the track **108.sub.2**.

(55) In one embodiment, the drum **802** may include a surface within two circular guides to keep the strap **116** on the surface within the drum **802**. The strap **116** may be installed to wrap around the drum **802** and through the slot **228** of the strap connection member **204**. Thus, the strap **116** may be installed without adding any twists to the strap **116** that could cause the strap **116** to be tangled or to get stuck in the slot **228** during movement of the door **102**.

(56) FIG. **9** illustrates a view of how the door stop mechanism **118** is coupled to the door **102**. For example, the bracket **202** may be coupled to an inner side of the door **102**. FIG. **9** illustrates a view of an exterior side **902** of the door (e.g., the side of the door that faces the outside of the building).

(57) The door stop mechanism **118** may be positioned such that the curved outer braking surface **210** contacts an exterior surface **904** of the first track **108.sub.1** and/or the second track **108.sub.2** when the brake **208** is in an engaged position.

(58) FIG. **10** illustrates an exploded view of another embodiment of the doorstop mechanism **1000**. In one embodiment, the door stop mechanism **1000** may include the bracket **202** that is similar to the bracket **202** of the door stop mechanism **118** illustrated in FIG. **4**. In one embodiment, the door stop mechanism **1000** may also include the strap connection member **204** similar to the door stop mechanism **118**.

(59) However, the door stop mechanism **1000** may include a body **1002** and a brake **1004** that are designed differently than the body **206** and the brake **208** of the door stop mechanism **118**. In one embodiment, the body **1002** may include a protruding member **1006**. The slot **226** of the strap connection member **204** may be fitted over the protruding member **1006**. The spring **230**, secured by the screw **250**, may interact with the body **1002** and the strap connection member **204** by connection to the protruding member **1006** and the protruding member **219**.

(60) In one embodiment, the body **1002** may also include slots **1008** and **1010**. The protruding member **218** may be inserted through the slot **1008** and the protruding member **219** may be inserted

through the slot **1010**.

(61) In one embodiment, the body **1002** may include a guide slot **1012**. The guide slot **1012** may interact with the brake **1004** to help guide movement of the brake **1004** between an engaged position and a disengaged position as the body **1002** moves up and down. For example, tension on the strap **116** may cause the body **1002** to move upwards and allow the guide slot **1012** to rotate the braking surface **1016** of the brake **1004** away from the exterior surface **904** of the track **108.sub.1** and/or track **108.sub.2**. When tension is lost on the strap **116**, the lost tension may cause the body **1002** to move downwards and allow the guide slot **1012** to rotate the braking surface **1016** of the brake **1004** towards the exterior surface **904** of the track **108.sub.1** and/or track **108.sub.2**.

(62) The guide slot **1012** may have a width that is equal to or slightly larger than a diameter of a protruding member **1030** on the brake **1004**. The guide slot **1012** may have a length that allows the brake **1004** to move between the engaged and disengaged position. The guide slot **1012** may be set at an angle relative to the vertical up and down movement of the body **1002** to translate a linear motion of the guide slot **1012** into a rotational movement of the brake **1004**.

(63) In one embodiment, the brake **1004** may include an opening **1020**. The brake **1004** may be positioned by inserting the protruding member **219** through the slot **1010** of the body **1002** and the opening **1020** of the brake **1004**. The protruding member **1030** on the brake **1004** may be inserted into the guide slot **1012**.

(64) A screw **1028** may be inserted into the protruding member **219** to secure the brake **1004** against the body **1002** and the bracket **202**. In one embodiment, one or more spacers **1022** and **1024** and a washer **1024** may be inserted between the opening **1020** and the screw **1028**.

(65) In one embodiment, the brake **1004** may include a braking surface **1016** that is similar to the braking surface **210**. For example, the braking surface **1016** may include a textured surface (e.g., teeth **1018.sub.1-1018.sub.n** illustrated in FIG. **10**) or any other type of textured surface such as a filed textured surface with a cross-hatch pattern.

(66) Thus, the present disclosure provides a mechanical door stop mechanism **118** or **1000** that can be deployed to stop movement of the door **102**. As noted above, although the door stop mechanism **118** and **1000** are shown on a single side of the bottom most panel **104.sub.1**, it should be noted that the door stop mechanism **118** or **1000** may be deployed on both sides or either side of the bottom most panel **104.sub.1**. Thus, the door stop mechanism **118** or **1000** may be deployed to interact with an outer or exterior surface **904** of the track **108.sub.1** and/or track **108.sub.2** where the overhead door system **100** is installed.

(67) It will be appreciated that variants of the above-disclosed and other features and functions, or alternatives thereof, may be combined into many other different systems or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

Claims

1. A door stop mechanism, comprising: a bracket comprising a first side and a second side coupled together at a right angle, wherein the first side comprises one or more openings to be connected to a bottom most panel of an overhead door and the second side comprising a first protruding member and a second protruding member; a body, comprising: a first slot to receive the first protruding member of the bracket; a second slot to receive the second protruding member of the bracket; and a third protruding member, wherein the third protruding member, the first slot, and the second slot are aligned in a vertical direction; a connection member, comprising a third slot to receive the third protruding member of the body; a spring coupled to the body and to the connection member; and a brake positioned adjacent to the body, wherein the brake moves between an engaged position via a spring force of the spring that causes the body to interact with the brake such that the brake

contacts a track of the overhead door and a disengaged position via a force that acts against the spring force to pull the body away from the brake such that the brake moves away from the track to allow the overhead door to move.

2. The door stop mechanism of claim 1, wherein the connection member further comprises: a slot to receive a strap.
3. The door stop mechanism of claim 2, wherein the spring is coupled to the third protruding member and the first protruding member.
4. The door stop mechanism of claim 1, the body further comprising: a cam surface to engage a back side of the brake, wherein movement of the cam surface causes the brake to rotate between the engaged position and the disengaged position.
5. The door stop mechanism of claim 1, wherein the brake comprises: a curved braking surface, wherein an outer side of the curved braking surface that faces the track of the overhead door comprises a plurality of contact points to contact the track.
6. The door stop mechanism of claim 5, wherein the plurality of contact points comprises teeth formed into the outer side of the curved braking surface.
7. The door stop mechanism of claim 1, wherein the brake is rotatably coupled via a torsion spring to the bracket.
8. The door stop mechanism of claim 1, wherein the body further comprises: a guide slot to receive a brake protruding member on the brake.
9. The door stop mechanism of claim 8, wherein the brake comprises: an opening to receive the second protruding member of the bracket; and the brake protruding member to be inserted into the guide slot of the body.
10. A door stop mechanism, comprising: a bracket comprising a first side and a second side coupled together at approximately 90 degrees, wherein the first side comprises one or more openings to be connected to a bottom most panel of an overhead door and the second side comprising a first protruding member and a second protruding member; a body, comprising: a first slot to receive the first protruding member of the bracket; a second slot to receive the second protruding member of the bracket; and a third protruding member, wherein the third protruding member the first slot, and the second slot are aligned in a vertical direction; a connection member, comprising a third slot to receive the third protruding member of the body; a spring coupled to the body and to the connection member; and a brake positioned adjacent to the body; a drum; and a strap coupled to the drum and to the connection member, wherein tension applied to the strap causes the body to move away from the brake to move to a disengaged position such that the brake moves away from a track of the overhead door, and a loss in tension to the strap causes the body to interact with the brake to move the brake to an engaged position such that the brake moves towards the track and contacts the track.
11. The door stop mechanism of claim 10, wherein a width of the strap is parallel to a width of the body and the connection member.
12. The door stop mechanism of claim 10, wherein the body further comprises: a guide slot.
13. The door stop mechanism of claim 12, wherein the connection member further comprises: a slot to receive the strap.
14. The door stop mechanism of claim 12, wherein the brake comprises: an opening to receive the second protruding member of the bracket; and a brake protruding member to be inserted into the guide slot of the body.
15. An overhead door system, comprising: a track system; an overhead door movably coupled to the track system, wherein the overhead door comprises a plurality of panels; a shaft located over the overhead door; a drum coupled to a first end of the shaft; a motor coupled to the first end of the shaft to rotate the shaft to open and close the overhead door and rotate the drum; a door stop mechanism coupled to a bottom most panel of the plurality of panels; a strap, wherein a first end of the strap is coupled to the drum and a second end of the strap is coupled to the door stop

mechanism; and a bracket coupled to the bottom most panel, wherein the bracket comprises a panel mounting surface and a door stop mechanism mounting surface coupled together perpendicularly, wherein the door stop mechanism mounting surface comprises a first protruding member and a second protruding member, wherein the door stop mechanism comprises: a body, comprising a first slot to receive the first protruding member of the bracket, a second slot to receive the second protruding member of the bracket, a third protruding member, and a guide slot, wherein the third protruding member, the first slot, and the second slot are aligned in a vertical direction; a strap connection member coupled to the second end of the strap, wherein the strap connection member comprises a third slot to receive the third protruding member of the body; a spring coupled to the body and the strap connection member; and a brake coupled to the second protruding member of the bracket and positioned adjacent to the body, wherein the brake comprises a brake protruding member that is inserted into the guide slot of the body, wherein tension applied to the strap causes the body to move away from the brake and causes the brake to rotate to a disengaged position such that the brake moves away from an exterior surface of the track system, and a loss in tension to the strap causes the body to move towards the brake and causes the brake to rotate to an engaged position such that the brake moves towards the track system and contacts the exterior surface of the track system.

16. The overhead door system of claim 15, wherein the brake comprises: a first end comprising an opening, wherein the second protruding member of the bracket is inserted through the opening; and a second end comprising a curved braking surface, wherein an outer side of the curved braking surface comprises a plurality of teeth to contact the track system.

17. The overhead door system of claim 16, wherein a screw is inserted through the opening of the first end of the brake and into the second protruding member to secure the brake against the body and the bracket.

18. The overhead door system of claim 15, wherein the guide slot is set at an angle relative to a vertical movement of the body to translate a linear motion of the guide slot into a rotational movement of the brake.
